

MATHEMATICS BASIC 2022 ALL SETS

=====

CHAPTER: 1 - Real Numbers

=====

YEAR: 2022
CHAPTER: 1 - Real Numbers
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 2
QUESTION: Which term of the A.P. 3, 8, 13, 18, ... is 78?
OPTIONS: (A) 14th term (B) 15th term (C) 16th term (D) 17th term
ANSWER: (C) 16th term

YEAR: 2022
CHAPTER: 1 - Real Numbers
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 2
QUESTION: Find the sum of the first twelve 2-digit multiples of 7, using an AP.
OPTIONS: (A) 630 (B) 588 (C) 756 (D) 672
ANSWER: (A) 630

YEAR: 2022
CHAPTER: 1 - Real Numbers
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 2
QUESTION: For an AP with common difference 6, the sum of first ten terms is same as four times the sum of first five terms. Determine the first term of the AP.
OPTIONS: (A) 3 (B) 4 (C) 5 (D) 6
ANSWER: (A) 3

YEAR: 2022
CHAPTER: 1 - Real Numbers
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 2
QUESTION: Determine the 36th term of the A.P. whose first two terms are -3 and 4 respectively.
OPTIONS: (A) 235 (B) 242 (C) 249 (D) 256
ANSWER: (B) 242

YEAR: 2022

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Write the next two terms of the A.P.: $\sqrt{27}$, $\sqrt{48}$, $\sqrt{75}$, ...

OPTIONS: (A) $\sqrt{96}$, $\sqrt{120}$ (B) $\sqrt{108}$, $\sqrt{147}$ (C) $\sqrt{100}$, $\sqrt{125}$ (D) $\sqrt{112}$, $\sqrt{148}$

ANSWER: (B) $\sqrt{108}$, $\sqrt{147}$

YEAR: 2022

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: In an AP, if $a = 50$, $d = -4$ and $S_n = 0$, then find the value of n .

OPTIONS: (A) 24 (B) 25 (C) 26 (D) 27

ANSWER: (C) 26

YEAR: 2022

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Which term of AP: $11/4$, $7/2$, $17/4$, ... is $53/4$?

OPTIONS: (A) 10th term (B) 11th term (C) 12th term (D) 13th term

ANSWER: (B) 11th term

YEAR: 2022

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: For an AP with common difference 6, the sum of first ten terms is same as four times the sum of first five terms. Determine the first term of the AP.

OPTIONS: (A) 3 (B) 4 (C) 5 (D) 6

ANSWER: (A) 3

YEAR: 2022

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: A solid metallic sphere of radius 3 cm is melted and recast into the shape of a solid cylinder of radius 2 cm. Find the height of the cylinder.

OPTIONS: (A) 6 cm (B) 7 cm (C) 8 cm (D) 9 cm

ANSWER: (D) 9 cm

YEAR: 2022

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: In an A.P., the first term is 12 and the common difference is 6. If the last term of the A.P. is 252, then find its middle term.

OPTIONS: (A) 122 (B) 132 (C) 142 (D) 152

ANSWER: (B) 132

YEAR: 2022

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: The sum of first n terms of an AP is given by $S_n = 3n^2 + 2n$. Find the AP.

OPTIONS: (A) 5, 11, 17, 23, ... (B) 5, 13, 21, 29, ... (C) 3, 9, 15, 21, ... (D) 7, 13, 19, 25, ...

ANSWER: (A) 5, 11, 17, 23, ...

=====
CHAPTER: 2 - Polynomials

=====

YEAR: 2022

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Find the value(s) of 'a' for which the quadratic equation $x^2 - ax + 1 = 0$ has real and equal roots.

OPTIONS: (A) $a = 1$ only (B) $a = -1$ only (C) $a = \pm 2$ (D) $a = 0$

ANSWER: (C) $a = \pm 2$

YEAR: 2022

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Find the nature of the roots of the quadratic equation $x^2 - 5x + 9 = 0$.

OPTIONS: (A) Real and equal (B) Real and distinct (C) No real roots (D) Cannot be determined

ANSWER: (C) No real roots

YEAR: 2022

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Write a quadratic equation with roots -3 and 5.

OPTIONS: (A) $x^2 - 2x - 15 = 0$ (B) $x^2 + 2x - 15 = 0$ (C) $x^2 - 8x + 15 = 0$ (D) $x^2 + 8x + 15 = 0$

ANSWER: (A) $x^2 - 2x - 15 = 0$

YEAR: 2022

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: If $x = 3$ is one root of the quadratic equation $2x^2 + px + 30 = 0$, find the value of p and the other root of the quadratic equation.

OPTIONS: (A) $p = -16$, other root = 5 (B) $p = -16$, other root = 3 (C) $p = 16$, other root = 5 (D) $p = 16$, other root = 3

ANSWER: (A) $p = -16$, other root = 5

YEAR: 2022

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: For what value of p , does the quadratic equation $px^2 + 2x + p = 0$ have real and equal roots?

OPTIONS: (A) $p = 1$ (B) $p = -1$ (C) $p = \pm 1$ (D) $p = 0$

ANSWER: (C) $p = \pm 1$

YEAR: 2022

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Solve the quadratic equation $2x^2 - 5x - 1 = 0$ for x .

OPTIONS: (A) $x = (5 + \sqrt{33})/4, (5 - \sqrt{33})/4$ (B) $x = (5 + \sqrt{17})/4, (5 - \sqrt{17})/4$ (C) $x = (5 + \sqrt{25})/4, (5 - \sqrt{25})/4$ (D) $x = (5 + \sqrt{41})/4, (5 - \sqrt{41})/4$

ANSWER: (A) $x = (5 + \sqrt{33})/4, (5 - \sqrt{33})/4$

YEAR: 2022

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Solve the quadratic equation $x^2 + 3x - 9 = 0$ for x .

OPTIONS: (A) $x = (-3 + \sqrt{45})/2, (-3 - \sqrt{45})/2$ (B) $x = (-3 + \sqrt{36})/2, (-3 - \sqrt{36})/2$ (C) $x = (-3 + \sqrt{27})/2, (-3 - \sqrt{27})/2$ (D) $x = (-3 + \sqrt{9})/2, (-3 - \sqrt{9})/2$

ANSWER: (A) $x = (-3 + \sqrt{45})/2, (-3 - \sqrt{45})/2$

YEAR: 2022

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Solve the quadratic equation $6 - x - x^2 = 0$ for x .

OPTIONS: (A) $x = 2, -3$ (B) $x = -2, 3$ (C) $x = 1, -6$ (D) $x = -1, 6$

ANSWER: (A) $x = 2, -3$

YEAR: 2022

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Find the value of 'k' so that the quadratic equation $3x^2 - 5x - 2k = 0$ has real and equal roots.

OPTIONS: (A) $k = -25/24$ (B) $k = 25/24$ (C) $k = -25/12$ (D) $k = 25/12$

ANSWER: (A) $k = -25/24$

YEAR: 2022

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Solve the equation $3x^2 - 8x - 1 = 0$ for x .

OPTIONS: (A) $x = (4 + \sqrt{19})/3, (4 - \sqrt{19})/3$ (B) $x = (8 + \sqrt{76})/6, (8 - \sqrt{76})/6$ (C) $x = (8 + \sqrt{64})/6, (8 - \sqrt{64})/6$ (D) $x = (4 + \sqrt{13})/3, (4 - \sqrt{13})/3$

ANSWER: (A) $x = (4 + \sqrt{19})/3, (4 - \sqrt{19})/3$

YEAR: 2022

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Solve for x : $2x^2 + 7/2 x + 3/4 = 0$.

OPTIONS: (A) $x = -3/2, -1/4$ (B) $x = -3/2, 1/4$ (C) $x = 3/2, -1/4$ (D) $x = 3/2, 1/4$

ANSWER: (A) $x = -3/2, -1/4$

YEAR: 2022

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: The sum of two numbers is 45. If 5 is subtracted from each of them, the product of these numbers becomes 124. Find the numbers.

OPTIONS: (A) 9 and 36 (B) 10 and 35 (C) 15 and 30 (D) 20 and 25

ANSWER: (A) 9 and 36

YEAR: 2022

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: A potter makes a certain number of pottery articles in a day. It was observed on a particular day the cost of production of each article (in rupees) was one more than twice the number of articles produced on that day. The total cost of production on that day was rupees 210. Taking number of articles produced on that day as x , form a quadratic equation in x .

OPTIONS: (A) $2x^2 + x - 210 = 0$ (B) $2x^2 - x - 210 = 0$ (C) $x^2 + 2x - 210 = 0$ (D) $x^2 - 2x - 210 = 0$

ANSWER: (A) $2x^2 + x - 210 = 0$

YEAR: 2022

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: The sum of the ages of a boy and his sister (in years) is 25 and product of their ages is 150. Find their present ages.

OPTIONS: (A) Boy 15, Sister 10 (B) Boy 12, Sister 13 (C) Boy 18, Sister 7 (D) Boy 20, Sister 5

ANSWER: (A) Boy 15, Sister 10

=====
CHAPTER: 3 - Pair of Linear Equations in Two Variables
=====

YEAR: 2022

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: The length of a rectangular park is 5 metres more than twice its breadth. If the area of the park is 250 sq m, find the length and breadth of the park.

OPTIONS: (A) Length 25 m, Breadth 10 m (B) Length 20 m, Breadth 12.5 m (C) Length 30 m, Breadth 8.33 m (D) Length 35 m, Breadth 7.14 m

ANSWER: (A) Length 25 m, Breadth 10 m

=====
CHAPTER: 4 - Quadratic Equations
=====

YEAR: 2022

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Find the value(s) of 'a' for which the quadratic equation $x^2 - ax + 1 = 0$ has real and equal roots.

OPTIONS: (A) $a = 1$ only (B) $a = -1$ only (C) $a = \pm 2$ (D) $a = 0$

ANSWER: (C) $a = \pm 2$

YEAR: 2022

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Find the nature of the roots of the quadratic equation $x^2 - 5x + 9 = 0$.

OPTIONS: (A) Real and equal (B) Real and distinct (C) No real roots (D) Cannot be determined

ANSWER: (C) No real roots

YEAR: 2022

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Write a quadratic equation with roots -3 and 5.

OPTIONS: (A) $x^2 - 2x - 15 = 0$ (B) $x^2 + 2x - 15 = 0$ (C) $x^2 - 8x + 15 = 0$ (D) $x^2 + 8x + 15 = 0$

ANSWER: (A) $x^2 - 2x - 15 = 0$

YEAR: 2022

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: If $x = 3$ is one root of the quadratic equation $2x^2 + px + 30 = 0$, find the value of p and the other root of the quadratic equation.

OPTIONS: (A) $p = -16$, other root = 5 (B) $p = -16$, other root = 3 (C) $p = 16$, other root = 5 (D) $p = 16$, other root = 3

ANSWER: (A) $p = -16$, other root = 5

YEAR: 2022

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: For what value of p , does the quadratic equation $px^2 + 2x + p = 0$ have real and equal roots?

OPTIONS: (A) $p = 1$ (B) $p = -1$ (C) $p = +1$ (D) $p = 0$

ANSWER: (C) $p = +1$

YEAR: 2022

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Solve the quadratic equation $2x^2 - 5x - 1 = 0$ for x .

OPTIONS: (A) $x = (5 + \sqrt{33})/4, (5 - \sqrt{33})/4$ (B) $x = (5 + \sqrt{17})/4, (5 - \sqrt{17})/4$ (C) $x = (5 + \sqrt{25})/4, (5 - \sqrt{25})/4$ (D) $x = (5 + \sqrt{41})/4, (5 - \sqrt{41})/4$

ANSWER: (A) $x = (5 + \sqrt{33})/4, (5 - \sqrt{33})/4$

YEAR: 2022

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Solve the quadratic equation $x^2 + 3x - 9 = 0$ for x .

OPTIONS: (A) $x = (-3 + \sqrt{45})/2, (-3 - \sqrt{45})/2$ (B) $x = (-3 + \sqrt{36})/2, (-3 - \sqrt{36})/2$ (C) $x = (-3 + \sqrt{27})/2, (-3 - \sqrt{27})/2$ (D) $x = (-3 + \sqrt{9})/2, (-3 - \sqrt{9})/2$

ANSWER: (A) $x = (-3 + \sqrt{45})/2, (-3 - \sqrt{45})/2$

YEAR: 2022

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Solve the quadratic equation $6 - x - x^2 = 0$ for x .

OPTIONS: (A) $x = 2, -3$ (B) $x = -2, 3$ (C) $x = 1, -6$ (D) $x = -1, 6$

ANSWER: (A) $x = 2, -3$

YEAR: 2022

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Find the value of ' k ' so that the quadratic equation $3x^2 - 5x - 2k = 0$ has real and equal roots.

OPTIONS: (A) $k = -25/24$ (B) $k = 25/24$ (C) $k = -25/12$ (D) $k = 25/12$

ANSWER: (A) $k = -25/24$

YEAR: 2022

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Solve the equation $3x^2 - 8x - 1 = 0$ for x .

OPTIONS: (A) $x = (4 + \sqrt{19})/3, (4 - \sqrt{19})/3$ (B) $x = (8 + \sqrt{76})/6, (8 - \sqrt{76})/6$ (C) $x = (8 + \sqrt{64})/6, (8 - \sqrt{64})/6$ (D) $x = (4 + \sqrt{13})/3, (4 - \sqrt{13})/3$

ANSWER: (A) $x = (4 + \sqrt{19})/3, (4 - \sqrt{19})/3$

YEAR: 2022

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Solve for x : $2x^2 + 7/2 x + 3/4 = 0$.

OPTIONS: (A) $x = -3/2, -1/4$ (B) $x = -3/2, 1/4$ (C) $x = 3/2, -1/4$ (D) $x = 3/2, 1/4$

ANSWER: (A) $x = -3/2, -1/4$

YEAR: 2022

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: The sum of two numbers is 45. If 5 is subtracted from each of them, the product of these numbers becomes 124. Find the numbers.

OPTIONS: (A) 9 and 36 (B) 10 and 35 (C) 15 and 30 (D) 20 and 25

ANSWER: (A) 9 and 36

YEAR: 2022

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: A potter makes a certain number of pottery articles in a day. It was observed on a particular day the cost of production of each article (in rupees) was one more than twice the number of articles produced on that day. The total cost of production on that day was rupees 210. Taking number of articles produced on that day as x , form a quadratic equation in x .

OPTIONS: (A) $2x^2 + x - 210 = 0$ (B) $2x^2 - x - 210 = 0$ (C) $x^2 + 2x - 210 = 0$ (D) $x^2 - 2x - 210 = 0$

ANSWER: (A) $2x^2 + x - 210 = 0$

YEAR: 2022

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: The sum of the ages of a boy and his sister (in years) is 25 and product of their ages is 150. Find their present ages.

OPTIONS: (A) Boy 15, Sister 10 (B) Boy 12, Sister 13 (C) Boy 18, Sister 7 (D) Boy 20, Sister 5

ANSWER: (A) Boy 15, Sister 10

=====

CHAPTER: 5 - Arithmetic Progressions

=====

YEAR: 2022

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Which term of the A.P. 3, 8, 13, 18, ... is 78?

OPTIONS: (A) 14th term (B) 15th term (C) 16th term (D) 17th term

ANSWER: (C) 16th term

YEAR: 2022

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Find the sum of the first twelve 2-digit multiples of 7, using an AP.

OPTIONS: (A) 630 (B) 588 (C) 756 (D) 672

ANSWER: (A) 630

YEAR: 2022

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: For an AP with common difference 6, the sum of first ten terms is same as four times the sum of first five terms. Determine the first term of the AP.

OPTIONS: (A) 3 (B) 4 (C) 5 (D) 6

ANSWER: (A) 3

YEAR: 2022

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Determine the 36th term of the A.P. whose first two terms are -3 and 4 respectively.

OPTIONS: (A) 235 (B) 242 (C) 249 (D) 256

ANSWER: (B) 242

YEAR: 2022

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Write the next two terms of the A.P.: $\sqrt{27}$, $\sqrt{48}$, $\sqrt{75}$, ...

OPTIONS: (A) $\sqrt{96}$, $\sqrt{120}$ (B) $\sqrt{108}$, $\sqrt{147}$ (C) $\sqrt{100}$, $\sqrt{125}$ (D) $\sqrt{112}$, $\sqrt{148}$

ANSWER: (B) $\sqrt{108}$, $\sqrt{147}$

YEAR: 2022

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: In an AP, if $a = 50$, $d = -4$ and $S_n = 0$, then find the value of n .

OPTIONS: (A) 24 (B) 25 (C) 26 (D) 27

ANSWER: (C) 26

YEAR: 2022

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Which term of AP: $\frac{11}{4}$, $\frac{7}{2}$, $\frac{17}{4}$, ... is $\frac{53}{4}$?

OPTIONS: (A) 10th term (B) 11th term (C) 12th term (D) 13th term

ANSWER: (B) 11th term

YEAR: 2022

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: For an AP with common difference 6, the sum of first ten terms is same as four times the sum of first five terms. Determine the first term of the AP.

OPTIONS: (A) 3 (B) 4 (C) 5 (D) 6

ANSWER: (A) 3

YEAR: 2022

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Which term of A.P. 3, 8, 13, 18, ... is 78?

OPTIONS: (A) 14th term (B) 15th term (C) 16th term (D) 17th term

ANSWER: (C) 16th term

YEAR: 2022

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Find the common difference of an A.P. whose n th term is given by $a_n = 6n - 5$.

OPTIONS: (A) 1 (B) 5 (C) 6 (D) 11

ANSWER: (C) 6

YEAR: 2022

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Find the sum of the first fifteen multiples of 8.

OPTIONS: (A) 840 (B) 960 (C) 1080 (D) 1200

ANSWER: (B) 960

YEAR: 2022

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Find the 8th term of an AP whose first term is -24 and 11th term is 21.

OPTIONS: (A) 3 (B) 6 (C) 9 (D) 12

ANSWER: (A) 3

YEAR: 2022

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Find the 15th term of an AP whose first term is 17 and fourth term is 44.

OPTIONS: (A) 134 (B) 140 (C) 146 (D) 152

ANSWER: (C) 146

YEAR: 2022

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: In an A.P., the first term is 12 and the common difference is 6. If the last term of the A.P. is 252, then find its middle term.

OPTIONS: (A) 122 (B) 132 (C) 142 (D) 152

ANSWER: (B) 132

YEAR: 2022

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: The sum of first n terms of an AP is given by $S_n = 3n^2 + 2n$. Find the AP.

OPTIONS: (A) 5, 11, 17, 23, ... (B) 5, 13, 21, 29, ... (C) 3, 9, 15, 21, ... (D) 7, 13, 19, 25, ...

ANSWER: (A) 5, 11, 17, 23, ...

YEAR: 2022

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: A pattern of stitches in circular rows is making a pattern: 6, 12, 18, 24, ... Check whether the given pattern forms an AP. If yes, find the common difference and the next term of the AP.

OPTIONS: (A) Yes, $d = 6$, next term = 30 (B) No, it does not form an AP (C) Yes, $d = 12$, next term = 36 (D) Yes, $d = 6$, next term = 36

ANSWER: (A) Yes, $d = 6$, next term = 30

YEAR: 2022

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: Write the n th term of the AP: 6, 12, 18, 24, ... Hence, find the number of stitches in the 10th circular row.

OPTIONS: (A) $a_n = 6n$, 10th term = 60 (B) $a_n = 6 + 6n$, 10th term = 66 (C) $a_n = 6n + 6$, 10th term = 66 (D) $a_n = 6n - 6$, 10th term = 54

ANSWER: (A) $a_n = 6n$, 10th term = 60

=====

CHAPTER: 6 - Coordinate Geometry

=====

YEAR: 2022

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: Draw a line segment of length 7 cm and divide it in the ratio 5:3.

OPTIONS: (A) Lengths are 4.375 cm and 2.625 cm (B) Lengths are 5 cm and 2 cm (C) Lengths are 4 cm and 3 cm (D) Lengths are 3.5 cm and 3.5 cm

ANSWER: (A) Lengths are 4.375 cm and 2.625 cm

YEAR: 2022

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: Draw a line segment of length 7.5 cm and divide it in the ratio 1:3.

OPTIONS: (A) Lengths are 1.875 cm and 5.625 cm (B) Lengths are 2 cm and 5.5 cm (C) Lengths are 2.5 cm and 5 cm (D) Lengths are 3 cm and 4.5 cm

ANSWER: (A) Lengths are 1.875 cm and 5.625 cm

YEAR: 2022

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: Draw a line segment of length 9.5 cm and divide it in the ratio 2:3.

OPTIONS: (A) Lengths are 3.8 cm and 5.7 cm (B) Lengths are 4 cm and 5.5 cm (C) Lengths are 3 cm and 6.5 cm (D) Lengths are 2 cm and 7.5 cm

ANSWER: (A) Lengths are 3.8 cm and 5.7 cm

=====

CHAPTER: 7 - Triangles

=====

YEAR: 2022

CHAPTER: 7 - Triangles

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: A circle is touching the side BC of triangle ABC at P and is touching AB and AC produced at Q and R respectively. Prove that $AQ = \frac{1}{2}$ (perimeter of triangle ABC).

OPTIONS: (A) $AQ = \frac{1}{2} (AB + BC + CA)$ (B) $AQ = \frac{1}{2} (AB + BC)$ (C) $AQ = \frac{1}{2} (AB + AC)$ (D) $AQ = \frac{1}{2} (BC + CA)$

ANSWER: (A) $AQ = \frac{1}{2} (AB + BC + CA)$

=====

CHAPTER: 8 - Circles

=====

YEAR: 2022

CHAPTER: 8 - Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: The distance between two tangents parallel to each other of a circle is 13 cm. Find the radius of the circle.

OPTIONS: (A) 6 cm (B) 6.5 cm (C) 13 cm (D) 26 cm

ANSWER: (B) 6.5 cm

YEAR: 2022

CHAPTER: 8 - Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: In Figure, if tangents PA and PB drawn from a point P to a circle with centre O, are inclined to each other at an angle of 70 degree, then find the measure of angle POA.

OPTIONS: (A) 35 degrees (B) 55 degrees (C) 70 degrees (D) 110 degrees

ANSWER: (B) 55 degrees

YEAR: 2022

CHAPTER: 8 - Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: If tangents PA and PB drawn from a point P to a circle with centre O, are inclined to each other at an angle of 70 degree, then find the measure of angle POA.

OPTIONS: (A) 35 degrees (B) 55 degrees (C) 70 degrees (D) 110 degrees

ANSWER: (B) 55 degrees

YEAR: 2022

CHAPTER: 8 - Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: In Figure, PA and PB are tangents to the circle with centre at O. If angle APB = 70 degree, then find m angle AQB.

OPTIONS: (A) 70 degrees (B) 110 degrees (C) 35 degrees (D) 55 degrees

ANSWER: (D) 55 degrees

YEAR: 2022

CHAPTER: 8 - Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: In Fig. 1, perimeter of triangle PQR is 20 cm. Find the length of tangent PA.

OPTIONS: (A) 5 cm (B) 10 cm (C) 15 cm (D) 20 cm

ANSWER: (B) 10 cm

YEAR: 2022

CHAPTER: 8 - Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: In Fig. 2, BC is tangent to the circle at point B of circle centred at O. BD is a chord of the circle so that angle BAD = 55 degree. Find m angle DBC.

OPTIONS: (A) 35 degrees (B) 55 degrees (C) 90 degrees (D) 45 degrees

ANSWER: (B) 55 degrees

YEAR: 2022

CHAPTER: 8 - Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: In Figure 2, PQ is a tangent to the circle centred at O such that angle PQB = 120 degree. Find the measure of angle QAB.

OPTIONS: (A) 30 degrees (B) 60 degrees (C) 90 degrees (D) 120 degrees

ANSWER: (A) 30 degrees

YEAR: 2022

CHAPTER: 8 - Circles

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: Prove that a parallelogram circumscribing a circle is a rhombus.

OPTIONS: (A) A parallelogram circumscribing a circle has all sides equal (B) A parallelogram circumscribing a circle has all angles equal (C) A parallelogram circumscribing a circle has diagonals equal (D) A parallelogram circumscribing a circle is a rectangle

ANSWER: (A) A parallelogram circumscribing a circle has all sides equal

YEAR: 2022

CHAPTER: 8 - Circles

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: Prove that the perpendicular at the point of contact to the tangent to a circle passes through the centre of the circle.

OPTIONS: (A) The perpendicular at the point of contact passes through the centre (B) The perpendicular at the point of contact is parallel to the radius (C) The perpendicular at the point of contact is tangent to the circle (D) The perpendicular at the point of contact bisects the chord

ANSWER: (A) The perpendicular at the point of contact passes through the centre

YEAR: 2022

CHAPTER: 8 - Circles

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: In Figure 3, the tangent l is parallel to the tangent m drawn at points A and B respectively to a circle centred at O . PQ is a tangent to the circle at R . Prove that angle $POQ = 90^\circ$.

OPTIONS: (A) angle $POQ = 90^\circ$ (B) angle $POQ = 45^\circ$ (C) angle $POQ = 60^\circ$ (D) angle $POQ = 120^\circ$

ANSWER: (A) angle $POQ = 90^\circ$

=====

CHAPTER: 9 - Introduction to Trigonometry

=====

YEAR: 2022

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: A solid metallic sphere of radius 3 cm is melted and recast into the shape of a solid cylinder of radius 2 cm. Find the height of the cylinder.

OPTIONS: (A) 6 cm (B) 7 cm (C) 8 cm (D) 9 cm

ANSWER: (D) 9 cm

=====

CHAPTER: 11 - Heights and Distances

=====

YEAR: 2022

CHAPTER: 11 - Heights and Distances

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: Two poles of heights 25 m and 35 m stand vertically on the ground. The tops of two poles are connected by a wire, which is inclined to the horizontal at an angle of 30° . Find the length of the wire and the distance between the poles.

OPTIONS: (A) Wire = 20 m, Distance = $10\sqrt{3}$ m (B) Wire = 20 m, Distance = 20 m (C) Wire = 40 m, Distance = $20\sqrt{3}$ m (D) Wire = 30 m, Distance = $15\sqrt{3}$ m

ANSWER: (A) Wire = 20 m, Distance = $10\sqrt{3}$ m

YEAR: 2022

CHAPTER: 11 - Heights and Distances

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: From a point on the ground, the angles of elevation of the bottom and the top of a transmission tower fixed at the top of a 20 m high building are 45 degree and 60 degree respectively. Find the height of the transmission tower.

OPTIONS: (A) $20(\sqrt{3} - 1)$ m (B) $20\sqrt{3}$ m (C) $20(\sqrt{3} + 1)$ m (D) $40\sqrt{3}$ m

ANSWER: (A) $20(\sqrt{3} - 1)$ m

YEAR: 2022

CHAPTER: 11 - Heights and Distances

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: From the top of a 7 m high building, the angle of elevation of the top of a cable tower is 60 degree and the angle of depression of its foot is 45 degree. Determine the height of the tower.

OPTIONS: (A) $7(\sqrt{3} + 1)$ m (B) $7\sqrt{3}$ m (C) 14 m (D) $7(\sqrt{3} - 1)$ m

ANSWER: (A) $7(\sqrt{3} + 1)$ m

YEAR: 2022

CHAPTER: 11 - Heights and Distances

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: Two poles of equal heights are standing opposite each other on either side of the road of width 60 m. From a point P between them on the road, the angles of elevation of the top of the poles are 60 degree and 30 degree respectively. Find the height of the poles.

OPTIONS: (A) $15\sqrt{3}$ m (B) $30\sqrt{3}$ m (C) $60\sqrt{3}$ m (D) $20\sqrt{3}$ m

ANSWER: (A) $15\sqrt{3}$ m

YEAR: 2022

CHAPTER: 11 - Heights and Distances

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: A tower stands vertically on the ground. From a point on the ground, which is 15 m away from the foot of the tower, the angle of elevation of the top of the tower is 60 degree. Find the height of the tower.

OPTIONS: (A) 15 m (B) $15\sqrt{3}$ m (C) 30 m (D) $30\sqrt{3}$ m

ANSWER: (B) $15\sqrt{3}$ m

YEAR: 2022

CHAPTER: 11 - Heights and Distances

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: In Figure 2, the angles of elevation of the top of a tower AB of height h m, from two points P and Q at a distance of x m and y m from the base of the tower respectively and in the same straight line with it, are 60 degree and 30 degree respectively. Prove that $h^2 = xy$.

OPTIONS: (A) $h^2 = xy$ (B) $h^2 = x/y$ (C) $h^2 = y/x$ (D) $h^2 = x + y$

ANSWER: (A) $h^2 = xy$

YEAR: 2022

CHAPTER: 11 - Heights and Distances

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: From a point on the ground, the angle of elevation of a vertical tower is found to be α such that $\tan \alpha = 1/3$. After walking 100 m towards the tower, the angle of elevation β becomes such that $\tan \beta = 3/4$. Find the height of the tower.

OPTIONS: (A) 60 m (B) 80 m (C) 100 m (D) 120 m

ANSWER: (A) 60 m

YEAR: 2022

CHAPTER: 11 - Heights and Distances

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: As observed from the top of a light house 100 m above sea level, the angle of depression of a ship, sailing directly towards it, changes from 30 degree to 45 degree.

Determine the distance travelled by the ship during this time. (Use $\sqrt{3} = 1.73$)

OPTIONS: (A) 73 m (B) 100 m (C) 173 m (D) 200 m

ANSWER: (A) 73 m

YEAR: 2022

CHAPTER: 11 - Heights and Distances

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: As observed from the top of a 100 m high lighthouse from the sea-level, the angles of depression of two ships are found to be 30 degree and 45 degree. If one ship is exactly behind the other on the same side of the lighthouse, find the distance between the two ships.

(Use $\sqrt{3} = 1.732$)

OPTIONS: (A) 73.2 m (B) 100 m (C) 173.2 m (D) 27.3 m

ANSWER: (A) 73.2 m

YEAR: 2022

CHAPTER: 11 - Heights and Distances

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: The angles of depression of the top and bottom of a 6 m tall building from the top of a multi-storeyed building are 30 degree and 45 degree respectively. Find the height of the multi-storeyed building and the distance between the two buildings. (Use $\sqrt{3} = 1.73$)

OPTIONS: (A) Height = 14.19 m, Distance = 8.19 m (B) Height = 12 m, Distance = 6 m (C)

Height = 18 m, Distance = 12 m (D) Height = 10 m, Distance = 7 m

ANSWER: (A) Height = 14.19 m, Distance = 8.19 m

YEAR: 2022

CHAPTER: 11 - Heights and Distances

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: A statue, 1.8 m tall, stands on the top of a pedestal. From a point on the ground, the angle of elevation of top of the statue is 60 degree and from the same point the angle of elevation of top of the pedestal is 45 degree. Find the height of the pedestal. (Use $\sqrt{3} = 1.73$)

OPTIONS: (A) 2.46 m (B) 1.8 m (C) 3.6 m (D) 4.2 m

ANSWER: (A) 2.46 m

YEAR: 2022

CHAPTER: 11 - Heights and Distances

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: The TV Tower stands vertically on the ground. From a point A on the ground, the angle of elevation of top of the tower (point B) is 60 degree. There is a point C on the tower which is 78 m (approx.) above the ground. The angle of elevation of the point C from point A is found to be 30 degree. Find the height of the tower and the distance of the tower from point A.

OPTIONS: (A) Height = 234 m, Distance = $78\sqrt{3}$ m (B) Height = 156 m, Distance = 78 m (C) Height = 234 m, Distance = 78 m (D) Height = 312 m, Distance = $78\sqrt{3}$ m

ANSWER: (A) Height = 234 m, Distance = $78\sqrt{3}$ m

YEAR: 2022

CHAPTER: 11 - Heights and Distances

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: The string of a kite is 100 metres long and it makes an angle of 60 degree with the horizontal. Find the height of the kite, assuming that there is no slack in the string. (Use $\sqrt{3} = 1.73$)

OPTIONS: (A) 86.5 m (B) 50 m (C) 173 m (D) 100 m

ANSWER: (A) 86.5 m

=====

CHAPTER: 12 - Areas Related to Circles

=====

YEAR: 2022

CHAPTER: 12 - Areas Related to Circles

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: A spherical glass vessel has a cylindrical neck 8 cm long and 1 cm in radius. The radius of the spherical part is 9 cm. Find the amount of water (in litres) it can hold, when filled completely.

OPTIONS: (A) 3.08 litres (B) 3.5 litres (C) 4.2 litres (D) 2.8 litres

ANSWER: (A) 3.08 litres

YEAR: 2022

CHAPTER: 12 - Areas Related to Circles

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: From a solid cylinder, whose height is 2.4 cm and diameter 1.4 cm a conical cavity of the same height and same diameter is hollowed out. Find the total surface area of the remaining solid.

OPTIONS: (A) 17.6 sq cm (B) 15.2 sq cm (C) 20.4 sq cm (D) 12.8 sq cm

ANSWER: (A) 17.6 sq cm

=====

CHAPTER: 13 - Surface Areas and Volumes

=====

YEAR: 2022

CHAPTER: 13 - Surface Areas and Volumes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: A solid metallic sphere of radius 3 cm is melted and recast into the shape of a solid cylinder of radius 2 cm. Find the height of the cylinder.

OPTIONS: (A) 6 cm (B) 7 cm (C) 8 cm (D) 9 cm

ANSWER: (D) 9 cm

YEAR: 2022

CHAPTER: 13 - Surface Areas and Volumes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: 3 cubes each of 8 cm edge are joined end to end. Find the total surface area of the cuboid so formed.

OPTIONS: (A) 896 sq cm (B) 768 sq cm (C) 1024 sq cm (D) 640 sq cm

ANSWER: (A) 896 sq cm

YEAR: 2022

CHAPTER: 13 - Surface Areas and Volumes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: 3 cubes each of 8 cm edge are joined end to end. Find the total surface area of the cuboid.

OPTIONS: (A) 896 sq cm (B) 768 sq cm (C) 1024 sq cm (D) 640 sq cm

ANSWER: (A) 896 sq cm

YEAR: 2022

CHAPTER: 13 - Surface Areas and Volumes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: A solid hemisphere of radius 8 cm is melted and recast into 4 identical right circular solid cones of base radius 4 cm. Find the height of the cone.

OPTIONS: (A) 8 cm (B) 12 cm (C) 16 cm (D) 20 cm

ANSWER: (C) 16 cm

YEAR: 2022

CHAPTER: 13 - Surface Areas and Volumes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: How many solid cones of height 3 cm and radius 2 cm can be formed by melting a solid sphere of radius 3 cm?

OPTIONS: (A) 3 (B) 6 (C) 9 (D) 12

ANSWER: (C) 9

YEAR: 2022

CHAPTER: 13 - Surface Areas and Volumes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: How many solid cones of height 3 cm and radius 2 cm can be formed by melting a solid sphere of radius 3 cm?

OPTIONS: (A) 3 (B) 6 (C) 9 (D) 12

ANSWER: (C) 9

YEAR: 2022

CHAPTER: 13 - Surface Areas and Volumes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: A metallic hollow cylindrical pipe has outer and inner radii as 6 cm and 4 cm respectively. Find the volume of the metal used in the pipe of length of 14 cm.

OPTIONS: (A) 880 cu cm (B) 440 cu cm (C) 1320 cu cm (D) 1760 cu cm

ANSWER: (A) 880 cu cm

YEAR: 2022

CHAPTER: 13 - Surface Areas and Volumes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Find the curved surface area of a right circular cone whose height is 15 cm and base radius is 8 cm. (Use $\pi = \frac{22}{7}$)

OPTIONS: (A) $\frac{2992}{7}$ sq cm (B) $\frac{3000}{7}$ sq cm (C) $\frac{2500}{7}$ sq cm (D) $\frac{2800}{7}$ sq cm

ANSWER: (A) $\frac{2992}{7}$ sq cm

YEAR: 2022

CHAPTER: 13 - Surface Areas and Volumes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: The surface area of a sphere is 616 sq cm. Find its radius. (Use $\pi = \frac{22}{7}$)

OPTIONS: (A) 7 cm (B) 14 cm (C) 21 cm (D) 28 cm

ANSWER: (A) 7 cm

YEAR: 2022

CHAPTER: 13 - Surface Areas and Volumes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: A cubical block of side 7 cm is surmounted by a hemisphere of largest possible diameter. Find the total surface area of the solid.

OPTIONS: (A) 332.5 sq cm (B) 350 sq cm (C) 400 sq cm (D) 450 sq cm

ANSWER: (A) 332.5 sq cm

YEAR: 2022

CHAPTER: 13 - Surface Areas and Volumes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: A solid is in the shape of a cone standing on a hemisphere with both their radii being equal to 1 cm and the height of the cone is equal to its radius. Find the volume of the solid in terms of π .

OPTIONS: (A) π cu cm (B) 2π cu cm (C) 3π cu cm (D) 4π cu cm

ANSWER: (A) π cu cm

YEAR: 2022

CHAPTER: 13 - Surface Areas and Volumes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Find the volume of a solid hemisphere whose total surface area is 462 sq m.

OPTIONS: (A) $2156/3$ cu m (B) 1000 cu m (C) 2156 cu m (D) 718.67 cu m

ANSWER: (A) $2156/3$ cu m

YEAR: 2022

CHAPTER: 13 - Surface Areas and Volumes

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: A company is supposed to make 50 toys in the form of a hemisphere surmounted by a right circular cone of the same base radius as that of hemisphere. If the radius of the base of the cone is 21 cm and height is 28 cm, find the volume of 50 toys.

OPTIONS: (A) 1617000 cu cm (B) 161700 cu cm (C) 16170000 cu cm (D) 16170 cu cm

ANSWER: (A) 1617000 cu cm

YEAR: 2022

CHAPTER: 13 - Surface Areas and Volumes

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: Find the ratio of the volume of hemisphere to the volume of cone.

OPTIONS: (A) 2:3 (B) 3:2 (C) 1:2 (D) 2:1

ANSWER: (A) 2:3

YEAR: 2022

CHAPTER: 13 - Surface Areas and Volumes

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: A solid cuboidal toy is made of wood. It has five cone shaped cavities to hold toy carrots. The dimensions of the toy are cuboid - 10 cm x 10 cm x 8 cm. Each cone carved out - Radius = 2.1 cm and Height = 6 cm. Find the volume of wood carved out to make five conical cavities.

OPTIONS: (A) 1386 cu cm (B) 138.6 cu cm (C) 13860 cu cm (D) 13.86 cu cm

ANSWER: (A) 1386 cu cm

YEAR: 2022

CHAPTER: 13 - Surface Areas and Volumes

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: Find the volume of the wood in the final product.

OPTIONS: (A) 6614 cu cm (B) 800 cu cm (C) 1386 cu cm (D) 6000 cu cm

ANSWER: (A) 6614 cu cm

YEAR: 2022

CHAPTER: 13 - Surface Areas and Volumes

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: Water standing on the roof is released into the cuboidal pit. If the cuboidal pit is filled completely by the roof water, then find the height of standing water on the roof.

OPTIONS: (A) 0.18 m (B) 1.8 m (C) 0.018 m (D) 18 m

ANSWER: (A) 0.18 m

YEAR: 2022

CHAPTER: 13 - Surface Areas and Volumes

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: If a cylindrical pit with diameter 3 m and height 2 m had been built, then which tank would hold more water?

OPTIONS: (A) Cuboidal pit holds more water (B) Cylindrical pit holds more water (C) Both hold equal water (D) Cannot be determined

ANSWER: (A) Cuboidal pit holds more water

=====

CHAPTER: 14 - Statistics

=====

YEAR: 2022

CHAPTER: 14 - Statistics

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Find the mode of the following frequency distribution: Class: 0-20, 20-40, 40-60, 60-80, 80-100; Frequency: 8, 7, 12, 5, 3
OPTIONS: (A) 48.3 (B) 50 (C) 45 (D) 52.5
ANSWER: (A) 48.3

YEAR: 2022
CHAPTER: 14 - Statistics
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 2
QUESTION: Find the mode of the following frequency distribution: Class: 20-30, 30-40, 40-50, 50-60, 60-70; Frequency: 25, 30, 45, 42, 35
OPTIONS: (A) 48.3 (B) 50 (C) 45 (D) 52.5
ANSWER: (A) 48.3

YEAR: 2022
CHAPTER: 14 - Statistics
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 2
QUESTION: Find the mode of the following frequency distribution: Class: 20-30, 30-40, 40-50, 50-60, 60-70; Frequency: 25, 30, 45, 42, 35
OPTIONS: (A) 48.3 (B) 50 (C) 45 (D) 52.5
ANSWER: (A) 48.3

YEAR: 2022
CHAPTER: 14 - Statistics
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 2
QUESTION: Find the modal class of the following distribution: Class: 0-20, 20-40, 40-60, 60-80, 80-100; Frequency: 11, 22, 19, 13, 7
OPTIONS: (A) 20-40 (B) 40-60 (C) 60-80 (D) 0-20
ANSWER: (A) 20-40

YEAR: 2022
CHAPTER: 14 - Statistics
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 2

QUESTION: Find the median class of the following distribution: Class: 0-20, 20-40, 40-60, 60-80, 80-100; Frequency: 11, 22, 19, 13, 7
OPTIONS: (A) 20-40 (B) 40-60 (C) 60-80 (D) 0-20
ANSWER: (B) 40-60

YEAR: 2022
CHAPTER: 14 - Statistics
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 2
QUESTION: Find the modal class of the following distribution: Class: 50-55, 55-60, 60-65, 65-70, 70-75, 75-80; Frequency: 10, 15, 8, 13, 9, 5
OPTIONS: (A) 50-55 (B) 55-60 (C) 60-65 (D) 65-70
ANSWER: (B) 55-60

YEAR: 2022
CHAPTER: 14 - Statistics
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 2
QUESTION: Find the median class of the following distribution: Class: 50-55, 55-60, 60-65, 65-70, 70-75, 75-80; Frequency: 10, 15, 8, 13, 9, 5
OPTIONS: (A) 50-55 (B) 55-60 (C) 60-65 (D) 65-70
ANSWER: (C) 60-65

YEAR: 2022
CHAPTER: 14 - Statistics
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 2
QUESTION: Find the sum of the lower limit of the modal class and upper limit of the median class for the following distribution: Class: 50-55, 55-60, 60-65, 65-70, 70-75, 75-80; Frequency: 10, 15, 8, 13, 9, 5
OPTIONS: (A) 115 (B) 120 (C) 125 (D) 130
ANSWER: (B) 120

YEAR: 2022
CHAPTER: 14 - Statistics
ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Using the empirical relationship between the three measures of central tendency, find the median of a distribution, whose mean is 169 and mode is 175.

OPTIONS: (A) 170 (B) 171 (C) 172 (D) 173

ANSWER: (B) 171

YEAR: 2022

CHAPTER: 14 - Statistics

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Mode = 3 Median - 2 Mean. If Mode = 175 and Mean = 169, find Median.

OPTIONS: (A) 170 (B) 171 (C) 172 (D) 173

ANSWER: (B) 171

YEAR: 2022

CHAPTER: 14 - Statistics

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Find the mode of the following frequency distribution: Class: 1-3, 3-5, 5-7, 7-9, 9-11, 11-13; Frequency: 20, 45, 80, 55, 40, 12

OPTIONS: (A) 5.17 (B) 6.17 (C) 7.17 (D) 8.17

ANSWER: (B) 6.17

YEAR: 2022

CHAPTER: 14 - Statistics

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: The frequency distribution given below shows the weight of 40 students of a class. Find the median weight of the students. Weight (in kg): 40-45, 45-50, 50-55, 55-60, 60-65, 65-70; Number of Students: 9, 5, 8, 9, 6, 3

OPTIONS: (A) 52.5 kg (B) 53.75 kg (C) 55 kg (D) 50 kg

ANSWER: (B) 53.75 kg

YEAR: 2022

CHAPTER: 14 - Statistics

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: Determine the median marks for the following data: Marks: 0-10, 10-20, 20-30, 30-40, 40-50; Number of Students: 3, 8, 15, 10, 8

OPTIONS: (A) 27.3 (B) 30 (C) 25 (D) 28.5

ANSWER: (A) 27.3

YEAR: 2022

CHAPTER: 14 - Statistics

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: The following table shows the age of patients admitted in a hospital during a particular week. Age (in years): 5-15, 15-25, 25-35, 35-45, 45-55, 55-65; Number of Patients: 5, 12, 20, 24, 15, 4. Find the mean age of the patients.

OPTIONS: (A) 35.5 (B) 36.5 (C) 34.5 (D) 37.5

ANSWER: (A) 35.5

YEAR: 2022

CHAPTER: 14 - Statistics

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: The mileage (km/l) of 50 cars was recorded by a dealer and tabulated as given below. Mileage (in km/l): 10-12, 12-14, 14-16, 16-18, 18-20; Number of Cars: 13, 18, 10, 7, 2. Find mean of the above distribution.

OPTIONS: (A) 13.68 (B) 14 (C) 15 (D) 16

ANSWER: (A) 13.68

YEAR: 2022

CHAPTER: 14 - Statistics

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: Determine median of the following frequency distribution: Class: 15-20, 20-25, 25-30, 30-35, 35-40, 40-45; Frequency: 8, 13, 21, 12, 5, 4

OPTIONS: (A) 27.5 (B) 28.5 (C) 26.5 (D) 29.5

ANSWER: (A) 27.5

YEAR: 2022

CHAPTER: 14 - Statistics

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: Determine median of the following frequency distribution: Class: 15-20, 20-25, 25-30, 30-35, 35-40, 40-45; Frequency: 8, 13, 21, 12, 5, 4

OPTIONS: (A) 27.5 (B) 28.5 (C) 26.5 (D) 29.5

ANSWER: (A) 27.5

YEAR: 2022

CHAPTER: 14 - Statistics

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: The median of following frequency distribution is 25. Find the value of x. Class: 0-10, 10-20, 20-30, 30-40, 40-50; Frequency: 6, 9, 10, 8, x

OPTIONS: (A) 7 (B) 8 (C) 9 (D) 10

ANSWER: (A) 7

YEAR: 2022

CHAPTER: 14 - Statistics

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: The median of following frequency distribution is 25. Find the value of x. Class: 0-10, 10-20, 20-30, 30-40, 40-50; Frequency: 6, 9, 10, 8, x

OPTIONS: (A) 7 (B) 8 (C) 9 (D) 10

ANSWER: (A) 7

YEAR: 2022

CHAPTER: 14 - Statistics

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: The median of the following frequency distribution is 35. Find the value of x. Class: 0-10, 10-20, 20-30, 30-40, 40-50; Frequency: 6, 3, x, 12, 19

OPTIONS: (A) 8 (B) 10 (C) 12 (D) 15

ANSWER: (B) 10

YEAR: 2022

CHAPTER: 14 - Statistics

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: Find mean of the following frequency distribution: Class: 0-20, 20-40, 40-60, 60-80, 80-100; Frequency: 6, 8, 5, 9, 7

OPTIONS: (A) 51.7 (B) 50 (C) 55 (D) 48

ANSWER: (A) 51.7

YEAR: 2022

CHAPTER: 14 - Statistics

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: Find the mean of the following frequency distribution: Class: 10-15, 15-20, 20-25, 25-30, 30-35; Frequency: 4, 10, 5, 6, 5

OPTIONS: (A) 22.5 (B) 23.5 (C) 24.5 (D) 21.5

ANSWER: (B) 23.5

YEAR: 2022

CHAPTER: 14 - Statistics

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: Find the mean of the following frequency distribution: Class: 10-15, 15-20, 20-25, 25-30, 30-35; Frequency: 4, 10, 5, 6, 5

OPTIONS: (A) 22.5 (B) 23.5 (C) 24.5 (D) 21.5

ANSWER: (B) 23.5

YEAR: 2022

CHAPTER: 14 - Statistics

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: Determine the mean of the following frequency distribution: Class: 100-110, 110-120, 120-130, 130-140, 140-150, 150-160; Frequency: 5, 4, 6, 8, 3, 4

OPTIONS: (A) 129 (B) 130 (C) 128 (D) 131

ANSWER: (A) 129

YEAR: 2022

CHAPTER: 14 - Statistics

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: The following data shows the distance travelled by 50 existing public transport buses in a day. Daily distance travelled (in km): 100-120, 120-140, 140-160, 160-180, 180-200; Number of buses: 12, 14, 8, 6, 10. Find the median distance travelled by a bus.

OPTIONS: (A) 138.57 km (B) 140 km (C) 135 km (D) 142 km

ANSWER: (A) 138.57 km

YEAR: 2022

CHAPTER: 14 - Statistics

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: The following data shows the distance travelled by 50 existing public transport buses in a day. Daily distance travelled (in km): 100-120, 120-140, 140-160, 160-180, 180-200; Number of buses: 12, 14, 8, 6, 10. Find the mean distance travelled by a bus.

OPTIONS: (A) 145.2 km (B) 140 km (C) 150 km (D) 135 km

ANSWER: (A) 145.2 km

YEAR: 2022

CHAPTER: 14 - Statistics

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: The following data shows the distance travelled by 50 existing public transport buses in a day. Daily distance travelled (in km): 100-120, 120-140, 140-160, 160-180, 180-200; Number of buses: 12, 14, 8, 6, 10. Find the median distance travelled by a bus.

OPTIONS: (A) 138.57 km (B) 140 km (C) 135 km (D) 142 km

ANSWER: (A) 138.57 km

YEAR: 2022

CHAPTER: 14 - Statistics

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: The following data shows the distance travelled by 50 existing public transport buses in a day. Daily distance travelled (in km): 100-120, 120-140, 140-160, 160-180, 180-200; Number of buses: 12, 14, 8, 6, 10. Find the mean distance travelled by a bus.

OPTIONS: (A) 145.2 km (B) 140 km (C) 150 km (D) 135 km

ANSWER: (A) 145.2 km

